

SURGICAL DELIVERY

In a Caesarean delivery, or Caesarean section, the baby emerges through an incision, or cut, made through the mother's skin, abdomen, and uterine muscles. These days this operation is very safe and leaves only a small scar on the mother's lower abdomen. Caesareans are performed for a number of reasons. The baby may be stuck in the birth canal and growing distressed or at risk from lack of oxygen.

Chorion membrane surrounds amnion

Placental blood vessels divide over surface

Umbilical cord joins baby at umbilicus (navel)

Umbilical vessels in tough jelly to protect vessels from damage



Curved blades fit around baby's head

Lock

Handles held by obstetric surgeon

Most babies are born headfirst

FORCEPS DELIVERY

Obstetrical forceps were invented in the late 16th century. Their rounded blades fit the curvature of the baby's head. They are fitted separately while the baby is in the uterus, then the blades are locked together. The forceps assist the mother's own uterine contractions to ease a stuck baby along the birth canal.

Obstetrical forceps from the 18th century

BABY'S MILK

The word mammal comes from the presence of mammary glands in the female of the species. These make milk, or lactate, to nourish the baby. The human mammary glands are in the breasts on the chest.

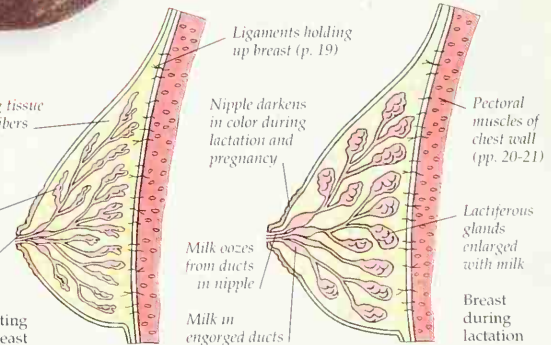
Each contains about 20 lobes, which are clusters of milk-making lactiferous glands. During pregnancy and after birth the glands enlarge and produce milk, which flows along lactiferous (milk) ducts and oozes from the nipple. When the baby sucks at the nipple, this stimulates milk to flow from the breasts.

Supporting tissue of fat and fibers

Lobe of lactiferous glands

Nipple

Non-lactating breast



Ligaments holding up breast (p. 19)

Nipple darkens in color during lactation and pregnancy

Milk oozes from ducts in nipple

Milk in engorged ducts

Pectoral muscles of chest wall (pp. 20-21)

Lactiferous glands enlarged with milk

Breast during lactation

Two umbilical arteries carry dark, low-oxygen blood
Umbilical vein carries bright red, high-oxygen blood

THE AFTERBIRTH

After the baby is born, the placenta comes away from the lining of the uterus, and it is delivered in the same way, which is why it is usually known as the "afterbirth." The baby is linked to the placenta by the umbilical cord, along which blood (pp. 32-33) flows in between baby and placenta. During pregnancy, the placenta brings together the baby's blood and the mother's blood, so that they are separated only by an extremely thin membrane. Oxygen and nutrients (p. 34) to help the baby grow pass from mother's blood to baby's blood, while the baby's wastes, like urea (pp. 38-39) and carbon dioxide, move in the opposite direction for disposal.